

Pragyan Chemical Plant — Maintenance Work Orders

Synthetic dataset generated for AI Knowledge Intelligence prototype. Equipment tags match the finalized P&ID set (PCP-PID-001 through PCP-PID-005). Note: formatting, naming conventions, and completeness are deliberately inconsistent across entries to simulate real-world messy maintenance records — this variation is intentional and useful for testing entity extraction/resolution.

WO-2024-0113

Date: 14-Mar-2024 **Equipment:** R-201 (Reactor) **Reported By:** Ramesh Iyer, Shift Engineer
Symptom: Jacket outlet temp reading 6°C higher than setpoint during batch 220. TIC-201 loop appears sluggish to respond. **Action Taken:** Checked CV-201 actuator — found partial fouling on valve seat restricting cooling water flow. Cleaned and re-calibrated. Verified TIC-201 response time back to normal (<90 sec). **Parts Used:** Valve seat gasket kit (1x)
Technician: Suresh Patil **Downtime:** 2.5 hrs **Status:** Closed

WO-2024-0114

Date: 15-Mar-2024 **Equipment:** P-101 **Reported:** Feed Pump A tripping intermittently on overload during startup. **Investigation/Action:** Motor amp draw checked — within spec. Suspected worn impeller bearing causing increased load. Switched over to P-102 (standby) per XV-101/XV-102 changeover procedure. P-101 pulled for bearing inspection. **Findings:** Bearing showed early-stage pitting, consistent with normal wear at ~14,000 operating hours. **Parts Used:** Bearing set (SKF 6205), lubricant **Tech:** S. Patil **Downtime:** 4 hrs (P-101 offline, P-102 covering) **Status:** Closed

WO-2024-0119

Date: 22-Mar-2024 **Equip Tag:** PSV-201 **Type:** Scheduled Test **Description:** Annual pop-test of reactor relief valve per OISD-STD-118 requirement. **Result:** Set pressure verified at 8.4 bar (spec: 8.0-8.5 bar). PASS. **Performed By:** Vikram Singh (Safety Officer), witnessed by M. Joshi **Next Due:** Mar-2025

WO-2024-0122

Date: 02-Apr-2024 **Equipment:** C-301 / Distillation Column **Complaint:** Operators reported unstable overhead pressure, PIC-301 hunting. **Root Cause Found:** PV-301 control valve

positioner drifting — output signal not tracking controller demand accurately. **Corrective Action:** Positioner recalibrated, stroke test performed 0-100%. Loop tuning parameters adjusted (reduced gain from 2.4 to 1.8). **Technician:** Meena Joshi (Instrumentation) **Downtime:** 1.5 hrs **Status:** Closed **Note:** Recommend positioner replacement at next turnaround — drift trend observed over last 3 work orders (see WO-2024-0087, WO-2024-0095).

WO-2024-0128

Date: 09-Apr-2024 **Equipment:** LV-301 (bottoms level control valve, C-301) Sticking intermittently, level oscillating +/- 8% around setpoint. Stem found partially seized. Disassembled, cleaned, re-lubricated stem packing. Functional test OK after rework. **Technician:** S. Patil **Parts:** Valve packing kit **Downtime:** 3 hrs **Status:** Closed

WO-2024-0134

Date: 18-Apr-2024 **Equipment:** V-401 (Raw Material Storage Tank) **Reported By:** D. Rao, Maintenance Supervisor **Issue:** Routine inspection flagged minor corrosion on LG-401 gauge glass fitting. **Action:** Fitting replaced, gauge glass inspected — no cracks found. Tank remained in service throughout (isolation not required for this repair). **Parts:** LG-401 fitting assembly **Technician:** R. Kulkarni **Downtime:** 0 hrs (no isolation needed) **Status:** Closed

WO-2024-0140

Date: 27-Apr-2024 **Equipment Tag:** XV-401 **Fault:** Valve handwheel stiff, difficult to operate during V-401 to P-401 transfer sequence. **Action:** Valve stem lubricated, gland packing adjusted. Operability confirmed — smooth operation restored. **Tech:** Suresh Patil **Downtime:** 45 min **Status:** Closed

WO-2024-0151

Date: 09-May-2024 **Equip:** P-401 Transfer Pump Unusual vibration noted during routine round by shift operator (A. Verma). Vibration analysis performed — bearing frequency spike detected on DE (drive end) bearing. Trend indicates early wear, not yet critical. **Recommendation:** Schedule bearing replacement at next planned maintenance window (within 60 days) — added to CMMS backlog as PM-2024-0038. **Technician:** M. Joshi **Status:** Monitoring / Open

WO-2024-0158

Date: 21-May-2024 **Equipment:** B-501 (Steam Boiler) **Priority:** HIGH **Reported By:** Ramesh Iyer **Symptom:** LIC-501 alarm — boiler drum low-level trip occurred during shift. Boiler auto-shutdown per safety interlock. No injuries, no damage. **Investigation:** LV-501 feed water control valve found stuck ~15% open due to scale buildup on valve trim, restricting adequate feed water flow to maintain drum level under load. **Action Taken:** LV-501 removed, trim cleaned and inspected, reinstalled. Feed water quality sampled — hardness slightly above spec, water treatment team notified separately (see related note WT-2024-014). **Parts:** None (cleaning only) **Technician:** Suresh Patil, assisted by D. Rao **Downtime:** 6 hrs (boiler offline for safe cooldown + repair) **Status:** Closed — RCA follow-up recommended on feed water quality

WO-2024-0163

Date: 30-May-2024 **Tag:** PSV-501 Annual relief valve test. Set pressure verified within spec (per OISD requirement). Witnessed: V. Singh. Result: PASS. Next due May-2025.

WO-2024-0170

Date: 08-Jun-2024 **Equipment:** FV-501 (fuel control valve, Boiler House) **Complaint:** FIC-501 loop unable to maintain stable fuel flow, boiler flame "hunting" observed by operator. **Diagnosis:** Valve actuator diaphragm found cracked, causing erratic positioning. **Action:** Actuator diaphragm replaced. Loop retuned. Flame stability confirmed over 2-hour observation period post-repair. **Parts Used:** Diaphragm kit (Fisher-type) **Technician:** M. Joshi **Downtime:** 3.5 hrs **Status:** Closed

WO-2024-0177

Date: 19-Jun-2024 **Equip:** ETP-601 Routine — chemical dosing pump for treatment tank found under-dosing due to partially blocked suction line. Line flushed, strainer cleaned. Dosing rate verified back to spec (per AIT-601 readings pre/post). Tech: S. Patil **Downtime:** 1 hr **Status:** Closed

WO-2024-0183

Date: 27-Jun-2024 **Equipment:** XV-603 (ETP discharge auto-close valve) **Priority:** HIGH — Safety Interlock Related **Reported:** Monthly interlock test — AAH-601 high alarm simulated, XV-603 auto-close function verified. Valve closed within 4 seconds of simulated

alarm (spec: <10 sec). PASS. **Note:** This is a safety-critical interlock per compliance requirement — test must be logged and retained for CPCB audit trail. **Performed By:** Vikram Singh, witnessed by D. Rao **Status:** Closed — Compliant

WO-2024-0191

Date: 05-Jul-2024 **Equipment:** P-601 (Effluent Transfer Pump) Seal leak observed during routine walkdown (minor weep, non-hazardous per MSDS review of effluent stream). Mechanical seal replaced as precaution. **Parts:** Mechanical seal kit **Technician:** R. Kulkarni **Downtime:** 2 hrs **Status:** Closed

WO-2024-0198

Date: 14-Jul-2024 **Equipment:** AGT-201 (Reactor Agitator Motor) **Reported By:** A. Verma, Process Engineer **Symptom:** Unusual motor hum, slight increase in current draw (batch 244 onward). **Action:** Motor bearing greased per schedule (was slightly overdue). Vibration check performed post-service — within normal range. Monitoring continued for next 3 batches to confirm resolution. **Technician:** M. Joshi **Downtime:** 0 hrs (serviced during planned batch changeover window) **Status:** Closed — Monitoring

WO-2024-0205

Date: 25-Jul-2024 **Equipment:** E-301 (Reboiler, C-301) Heat transfer efficiency degraded — TIC-301 requiring higher steam flow than historical baseline to maintain column bottoms temperature, suggesting possible fouling on tube side. **Action:** Scheduled for chemical cleaning at next planned shutdown (added to turnaround scope TA-2024-Q3). Interim: TV-301 setpoint adjusted to compensate. **Technician:** S. Patil / Engineering Review: A. Verma **Status:** Open — Deferred to Turnaround

WO-2024-0212

Date: 03-Aug-2024 **Equip Tag:** HS-101 / P-101 Hand switch (HOA) for Feed Pump A found in "OFF" position unexpectedly during shift changeover — cause unclear, possibly inadvertent operation during nearby work. No process impact (P-102 was running). Switch function tested normal, no fault found. Logged as anomaly for awareness. **Reported By:** R. Iyer **Status:** Closed — No Fault Found (NFF)

Summary

- **Total Work Orders:** 18
- **Equipment covered:** R-201, P-101, P-102, AGT-201, TIC-201, PSV-201, CV-201, HS-101, C-301, PIC-301, PV-301, LV-301, TIC-301, TV-301, E-301, V-401, LG-401, XV-401, P-401, B-501, LIC-501, LV-501, PSV-501, FIC-501, FV-501, ETP-601, XV-603, AAH-601, P-601
- **Open/Monitoring items:** WO-2024-0151 (P-401 bearing), WO-2024-0205 (E-301 fouling, deferred to turnaround)
- **Safety-critical/compliance-linked:** WO-2024-0119, WO-2024-0163 (PSV tests), WO-2024-0183 (XV-603 interlock test)
- Note the deliberate naming inconsistencies across entries (e.g., "P-101" vs "Feed Pump A", "R-201" vs "the reactor", "XV-603" tag sometimes given full context, sometimes abbreviated) — useful for testing your entity resolution/normalization layer.